

Exp-7

```
package servletDemo;
```

```
import java.net.*;
```

```
public class InetDemo {
```

```
    public static void main(String[] args) {
```

```
        try {
```

```
            InetAddress local = InetAddress.getLocalHost();
```

```
            System.out.println("Local Host:");
```

```
            System.out.println("Host Name: " + local.getHostName());
```

```
            System.out.println("IP Address: " + local.getHostAddress());
```

```
            InetAddress google = InetAddress.getByName("www.google.com");
```

```
            System.out.println("\nUsing getByName():");
```

```
            System.out.println("Host Name: " + google.getHostName());
```

```
            System.out.println("IP Address: " + google.getHostAddress());
```

```
            InetAddress[] addresses = InetAddress.getAllByName("www.google.com");
```

```
            System.out.println("\nUsing getAllByName():");
```

```
            for (InetAddress addr : addresses) {
```

```
                System.out.println(addr);
```

```
            }
```

```
            byte[] ip = {8, 8, 8, 8};
```

```
            InetAddress addr = InetAddress.getByAddress(ip);
```

```
            System.out.println("\nUsing getByAddress():");
```

```
            System.out.println(addr);
```

```
        } catch (Exception e) {
```

```
            System.out.println(e);
```

```
        }
```

```
    }
```

```
}
```

```
Problems Javadoc Declaration Console X
<terminated> InetDemo [Java Application] F:\Downloads\eclipse-jee-2022-06-R-win32-x86_64\eclipse\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_17.0.3.v20220515-1416\jre\bin\javaw.exe (26-Apr-2022)

Local Host:
Host Name: Hariom-Gugale1788
IP Address: 10.197.28.48

Using getByName():
Host Name: www.google.com
IP Address: 142.251.152.119

Using getAllByName():
www.google.com/142.251.152.119
www.google.com/142.251.156.119
www.google.com/142.251.157.119
www.google.com/142.251.155.119
www.google.com/142.251.154.119
www.google.com/142.251.153.119
www.google.com/142.251.151.119
www.google.com/142.251.150.119
www.google.com/2001:4860:482c:7700:0:0:0:0
www.google.com/2001:4860:4826:7700:0:0:0:0
www.google.com/2001:4860:4829:7700:0:0:0:0
www.google.com/2001:4860:482b:7700:0:0:0:0
www.google.com/2001:4860:482a:7700:0:0:0:0
www.google.com/2001:4860:4828:7700:0:0:0:0
www.google.com/2001:4860:482d:7700:0:0:0:0
www.google.com/2001:4860:4827:7700:0:0:0:0

Using getByAddress():
/8.8.8.8
```

Exp-8A

LoginServlet.java

```
package servletDemo;
```

```
import java.io.*;
```

```
import javax.servlet.*;
```

```
import javax.servlet.http.*;
```

```
import javax.servlet.annotation.WebServlet;
```

```
@WebServlet("/login")
```

```
public class LoginServlet extends HttpServlet {
```

```
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
```

```
        throws ServletException, IOException {
```

```
        response.setContentType("text/html");
```

```
        PrintWriter out = response.getWriter();
```

```
        // Get form data
```

```
        String username = request.getParameter("username");
```

```
        String password = request.getParameter("password");
```

```
        // Simple authentication (hardcoded)
```

```
        if(username.equals("Hariom") && password.equals("1234")) {
```

```
            out.println("<h2>Login Successful!</h2>");
```

```
            out.println("<h3>Welcome " + username + "</h3>");
```

```
        } else {
```

```
            out.println("<h2>Invalid Username or Password</h2>");
```

```
        }
```

```
    }
```

```
}
```

Login.html

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Login Form</title>
```

```
</head>
```

```
<body>
```

```
  <h2>Login Form</h2>
```

```
  <form action="login" method="post">
```

```
    Username: <input type="text" name="username"><br><br>
```

```
    Password: <input type="password" name="password"><br><br>
```

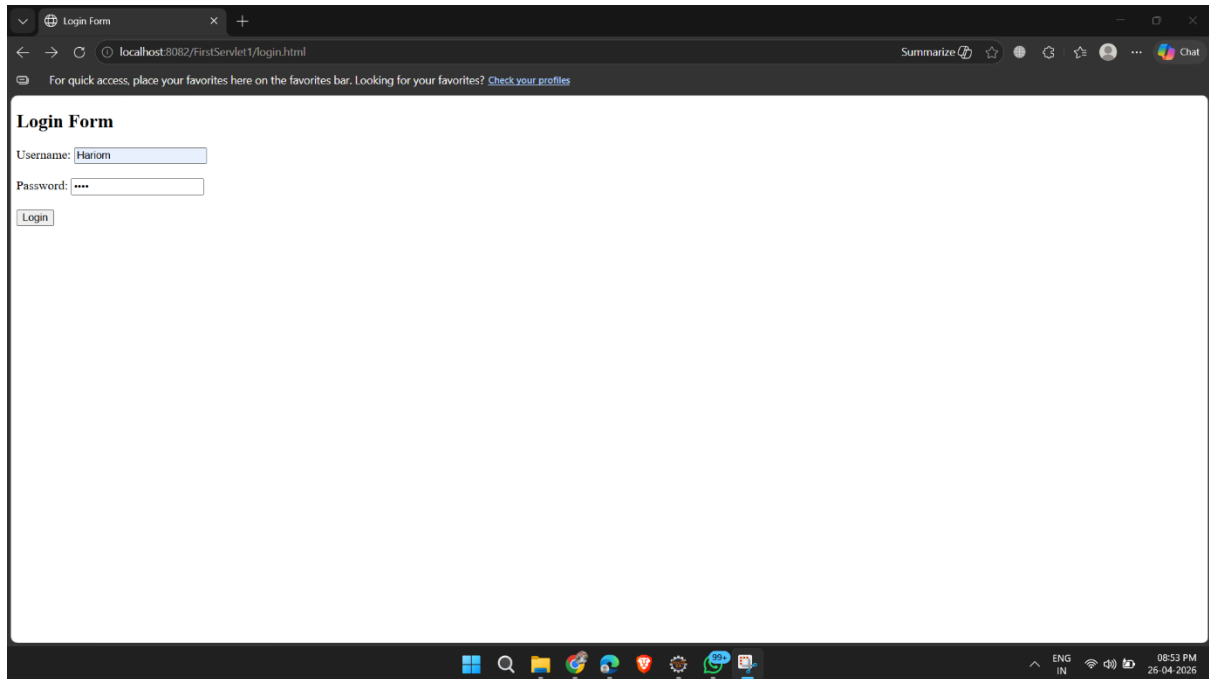
```
    <input type="submit" value="Login">
```

```
  </form>
```

```
</body>
```

```
</html>
```

Output



Exp-8B

CalculatorServlet.java

```
package servletDemo;
```

```
import java.io.*;
```

```
import javax.servlet.*;
```

```
import javax.servlet.http.*;
```

```
import javax.servlet.annotation.WebServlet;
```

```
@WebServlet("/calc")
```

```
public class CalculatorServlet extends HttpServlet {
```

```
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
```

```
        throws ServletException, IOException {
```

```
        response.setContentType("text/html");
```

```
        PrintWriter out = response.getWriter();
```

```
        double num1 = Double.parseDouble(request.getParameter("num1"));
```

```
        double num2 = Double.parseDouble(request.getParameter("num2"));
```

```
        String operation = request.getParameter("operation");
```

```
        double result = 0;
```

```
        if(operation.equals("Add")) {
```

```
            result = num1 + num2;
```

```
        }
```

```
        else if(operation.equals("Subtract")) {
```

```
            result = num1 - num2;
```

```

    }
    else if(operation.equals("Multiply")) {
        result = num1 * num2;
    }
    else if(operation.equals("Divide")) {
        if(num2 != 0)
            result = num1 / num2;
        else {
            out.println("<h3>Cannot divide by zero</h3>");
            return;
        }
    }

    out.println("<h2>Result = " + result + "</h2>");
}
}

```

Calculator.html

```

<!DOCTYPE html>

<html>
<head>
    <title>Simple Calculator</title>
</head>
<body>

    <h2>Simple Calculator</h2>

    <form action="calc" method="post">
        Enter Number 1: <input type="text" name="num1"><br><br>
        Enter Number 2: <input type="text" name="num2"><br><br>
    </form>

```

```
<input type="submit" name="operation" value="Add">  
<input type="submit" name="operation" value="Subtract">  
<input type="submit" name="operation" value="Multiply">  
<input type="submit" name="operation" value="Divide">
```

```
</form>
```

```
</body>
```

```
</html>
```


Output

